

Experiment 6

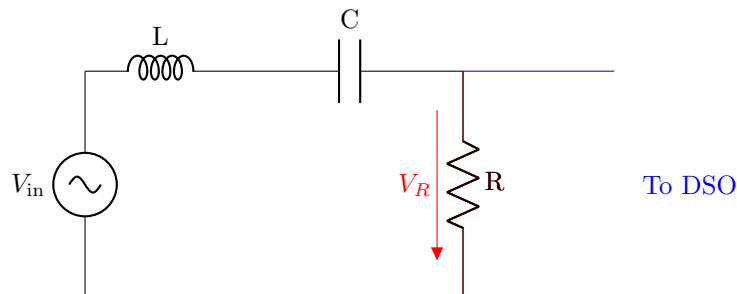
Steady-state response of RLC circuits

Components/Instruments/Meters

- Digital storage oscilloscope (DSO)
- Function generator
- Breadboard
- Resistor
- Capacitor
- Inductor
- Connecting wires

Frequency response of RLC circuit

1. Setup a series RLC circuit.



R	C	L
10 k Ω	220 nF	2 H

2. Set the signal generator to sine wave signal (Peak-to-peak value = 2 V, frequency = 10 Hz).
3. Connect 'Channel-1' of DSO to input terminals of the circuit.
4. Connect 'Channel-2' of DSO to observe the voltage across the resistor, i.e., V_R .

5. Vary the frequency of the input signal (till 1000 Hz) and measure the amplitude of V_R and the phase difference between V_{in} and V_R for different values of frequency.
6. Does the shape of the pattern in X-Y mode change as you vary the frequency?
7. Find the (resonant) frequency at which the voltage across the resistor is maximum (or the phase difference between the voltages is zero). Be as precise as you can. Also, identify the frequencies at which the amplitude is half of the maximum. What is the phase difference at these frequencies?
8. Plot the current and phase difference as a function of frequency.
9. Are you able to verify KVL at resonant frequency?
10. Are you able to verify KVL at any frequency, other than the resonant frequency?

Optional: What happens to the voltage/current at the resonance if $R = 1 \text{ k}\Omega$?